FOR A PERFECT MORPHISM THE SHRIEK OF THE STRUCTURE SHEAF IS BOUNDED COHERENT, OF FINITE TOR DIMENSION, SELF DUAL, PRESERVES BOUNDED COHERENT OBJECTS AND MAKES THE COMPARISON MAP AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0B6U (sources/stacks-duality/pinned-original-source.tex line 4798).

Allowed standard prerequisites (editorial).

- The upper shriek of a closed immersion, of an open immersion and of the affine line projection. Same chapter, Lemmas labelled lemma-shriek-closed-immersion, lemma-shriek-open-immersion and lemma-shriek-affine-line, cited at proof lines 4844-4846, 4884-4886 and 4900-4902. Compute f^! in each of the model cases used in the reduction.
- The comparison map mu and its compatibility with composition. Same chapter, Lemma labelled lemma-map-pullback-to-shriek-well-defined, invoked in the statement and cited at proof lines 4862-4868. Supplies the map whose bijectivity is assertion (e) and the diagram closing (e) under composition.
- Cohomological bounds for the shriek of coherent complexes and the perfect proper case. Same chapter, Lemma labelled lemma-shriek-coherent, cited at proof lines 4840-4842, and Lemma labelled lemma-compare-with-pullback-flat-proper-noetherian, cited at proof lines 4852-4854. Give the first preliminary remark and settle (e) for a perfect proper morphism.
- The local algebraic computation of the relative dualizing complex. Same chapter, Remark labelled remark-local-calculation-shriek and Dualizing Complexes, Lemma labelled dualizing-lemma-relative-dualizing-algebraic, cited at proof lines 4828-4832. Proves (a), (b) and (c) in the affine case.

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- Finiteness and perfection permanence for the factorization. Morphisms, Lemma labelled morphisms-lemma-quasi-affine-finite-type-over-S, cited at proof lines 4874-4876, and More on Morphisms, Lemmas labelled more-morphisms-lemma-flat-finite-presentation-perfect and more-morphisms-lemma-perfect-permanence, cited at proof lines 4819-4821 and 4878-4880. Produce the factorization into a perfect closed immersion and a projection.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6T: sources/stacks-duality/pinned-original-source.tex, source lines 4427–4427. Same-chapter pointer into the author environment lemma at source line 4427; invoked at proof line 4855 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 068A: sources/stacks-chow/tags.txt, source lines 7999–7999. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4823 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0BV2: sources/stacks-duality/pinned-original-source.tex, source lines 4636–4636. Same-chapter pointer into the author environment remark at source line 4636; invoked at proof line 4826 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BZW: sources/stacks-chow/tags.txt, source lines 15066–15066. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4828 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0DHY: sources/stacks-chow/tags.txt, source lines 16923–16923. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4830 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08E5: sources/stacks-chow/tags.txt, source lines 10661–10661. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4831 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08E8: sources/stacks-chow/tags.txt, source lines 10664–10664. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4832 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4833 of the counted proof. It is

retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AU1: sources/stacks-duality/pinned-original-source.tex, source lines 4686–4686. Same-chapter pointer into the author environment lemma at source line 4686; invoked at proof line 4840 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AU0: sources/stacks-duality/pinned-original-source.tex, source lines 4568–4568. Same-chapter pointer into the author environment lemma at source line 4568; invoked at proof line 4843 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9U: sources/stacks-duality/pinned-original-source.tex, source lines 3306–3306. Same-chapter pointer into the author environment lemma at source line 3306; invoked at proof line 4845 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 04II: sources/stacks-chow/tags.txt, source lines 5904–5904. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4863 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 068D: sources/stacks-chow/tags.txt, source lines 8002–8002. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4867 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA2: sources/stacks-duality/pinned-original-source.tex, source lines 4624–4624. Same-chapter pointer into the author environment lemma at source line 4624; invoked at proof line 4875 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA1: sources/stacks-duality/pinned-original-source.tex, source lines 4609–4609. Same-chapter pointer into the author environment lemma at source line 4609; invoked at proof line 4883 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. Assume f is perfect (e.g., flat). Then*

(a) *$f^!\mathcal{O}_Y$ is in $D^b_{Coh}(\mathcal{O}_X)$,*
(b) *$f^!\mathcal{O}_Y$ has finite tor dimension in $D(f^{-1}\mathcal{O}_Y)$,*
(c) *$\mathcal{O}_X \to R\mathcal{H}om_{\mathcal{O}_X}(f^!\mathcal{O}_Y, f^!\mathcal{O}_Y)$ is an isomorphism,*
(d) *$f^!$ maps $D^b_{Coh}(\mathcal{O}_Y)$ into $D^b_{Coh}(\mathcal{O}_X)$,*
(e) *the map $\mu_{f,K} : Lf^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} f^!\mathcal{O}_Y \to f^!K$ of Lemma 0B6T is an isomorphism for all $K \in D^+_{QCoh}(\mathcal{O}_Y)$.*

Proof. (A flat morphism of finite presentation is perfect, see More on Morphisms, Lemma 068A.) Assertions (a), (b), and (c) are local on X. Thus we may assume X and Y are affine. Then Remark 0BV2 turns (a), (b), and (c) into (1)(a), (1)(b),

and (1)(c) of Dualizing Complexes, Lemma 0BZW. (Use Derived Categories of Schemes, Lemmas 0DHY, 08E5, 08E8 0A6H to match the assertions.)

To prove (d) and (e) we begin with a series of preliminary remarks.

(1) We already know that $f^!$ sends $D^+_{Coh}(\mathcal{O}_Y)$ into $D^+_{Coh}(\mathcal{O}_X)$, see Lemma 0AU1.
(2) If f is an open immersion, then $f^! = f^*$ and (d) and (e) are true because we can take $\overline{X} = Y$ in the construction of $f^!$ and μ_f, see Lemma 0AU0.
(3) If f is a perfect proper morphism, then (e) is true by Lemma 0A9U.
(4) If there exists an open covering $X = \bigcup U_i$ and (d) is true for $U_i \to Y$, then (d) is true for $X \to Y$. Same for (e). This holds because the construction of $f^!$ and μ_f commutes with passing to open subschemes.
(5) If $g : Y \to Z$ is a second perfect morphism in FTS_S and (e) holds for f and g, then $f^!g^!\mathcal{O}_Z = Lf^*g^!\mathcal{O}_Z \otimes^{\mathbf{L}}_{\mathcal{O}_X} f^!\mathcal{O}_Y$ and (e) holds for $g \circ f$ by the commutative diagram of Lemma 0B6T.
(6) If (d) and (e) hold for both f and g, then (d) and (e) hold for $g \circ f$. Namely, then $f^!g^!\mathcal{O}_Z$ is bounded above (by the previous point) and $L(g \circ f)^*$ has finite cohomological dimension and (d) follows from (e) which we saw above.

From these points we see it suffices to prove (d) and (e) in case X is affine. Choose an immersion $X \to \mathbf{A}^n_Y$ (Morphisms, Lemma 04II) which we factor as $X \to U \to \mathbf{A}^n_Y \to Y$ where $X \to U$ is a closed immersion and $U \subset \mathbf{A}^n_Y$ is open. Note that $X \to U$ is a perfect closed immersion by More on Morphisms, Lemma 068D. Thus it suffices to prove the lemma for a perfect closed immersion and for the projection $\mathbf{A}^n_Y \to Y$.

Let $f : X \to Y$ be a perfect closed immersion. We already know (d) holds. Let $K \in D^b_{Coh}(\mathcal{O}_Y)$. Then $f^!K = R\mathcal{H}om(\mathcal{O}_X, K)$ (Lemma 0AA2) and $f_*f^!K = R\mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{O}_X, K)$. Since f is perfect, the complex $f_*\mathcal{O}_X$ is perfect and hence $R\mathcal{H}om_{\mathcal{O}_Y}(f_*\mathcal{O}_X, K)$ is bounded above. This proves that (d) holds. Some details omitted.

Let $f : \mathbf{A}^n_Y \to Y$ be the projection. Then (d) holds by repeated application of Lemma 0AA1. Finally, (e) is true because it holds for $\mathbf{P}^n_Y \to Y$ (flat and proper) and because $\mathbf{A}^n_Y \subset \mathbf{P}^n_Y$ is an open. □